

Ans - 1 :- Let the n consecutive integers be
 $a, a+1, a+2, \dots, a+(n-1)$

Now look at their remainders modulo n :
 $a \bmod n, (a+1) \bmod n, \dots, (a+n-1) \bmod n$

These n integers give n different remainders:
 $0, 1, 2, \dots, n-1$.

So among them, exactly one remainder must be 0.

Hence, among any n consecutive integers, exactly one is divisible by n .

Ans - 2 :- First eight positive integers are:
 $1, 2, 3, 4, 5, 6, 7, 8$

Group them into pairs whose sum is 9
 $(1, 8), (2, 7), (3, 6), (4, 5)$

So there are only 4 groups.

by pigeonhole formula: $\left\lceil \frac{N}{k} \right\rceil$

$N = \text{no. of objects} = 5$

$k = \text{no. of boxes} = 4$

$$\left\lceil \frac{5}{4} \right\rceil = \lceil 1.25 \rceil = 2$$

Hence, at least one pair contains two selected integers, so their sum is 9.

Ans-3 Given

$$a_n = 5a_{n-1} - 6a_{n-2}, \quad n \geq 2, \quad a_0 = 1, \quad a_1 = 0$$

Let

$$A(x) = \sum_{n=0}^{\infty} a_n x^n$$

Multiply the recurrence by x^n and sum from $n=2$ to ∞ :

$$\sum_{n=2}^{\infty} a_n x^n = 5 \sum_{n=2}^{\infty} a_{n-1} x^n - 6 \sum_{n=2}^{\infty} a_{n-2} x^n$$

Now simplify:

$$\sum_{n=2}^{\infty} a_n x^n = A(x) - a_1 - a_0 = A(x) - 1$$

$$\sum_{n=2}^{\infty} a_{n-1} x^n = x(A(x) - 1)$$

$$[\because a_0 = 1 \text{ \& } a_1 = 0]$$

$$\sum_{n=2}^{\infty} a_{n-2} x^n = x^2 A(x)$$

So,

$$A(x) - 1 = 5x(A(x) - 1) - 6x^2 A(x)$$

$$A(x)(1 - 5x + 6x^2) = 1 - 5x$$

$$A(x) = \frac{1 - 5x}{1 - 5x + 6x^2}$$

Ans 4:-

Given

$$a_n = a_{n-2}, \quad n \geq 2, \quad a_0 = 5, \quad a_1 = -1$$

let $A(x) = \sum_{n=0}^{\infty} a_n x^n$

Multiplying x^2 and sum from 2 to ∞ :

$$\sum_{n=2}^{\infty} a_n x^n = \sum_{n=2}^{\infty} a_{n-2} x^n$$

Now simplify:

$$\sum_{n=2}^{\infty} a_n x^n = A(x) - 5 + x \quad [\because a_0 = 5 \text{ \& } a_1 = -1]$$

$$\sum_{n=2}^{\infty} a_{n-2} x^n = x^2 A(x)$$

So,

$$A(x) - 5 + x = x^2 A(x)$$

$$A(x) = \frac{5-x}{1-x^2}$$

Ans-5:- The generating function is:

$$G(x) = 1 + 4x + 16x^2 + 64x^3 + 256x^4$$

This is a finite geometric-type expression with ratio $4x$.
using formula

$$1 + r + r^2 + \dots + r^n = \frac{1 - r^{n+1}}{1 - r}$$

with $r = 4x$, we get

$$G(x) = 1 + 4x + (4x)^2 + (4x)^3 + (4x)^4$$

Therefore,
$$G(x) = \frac{1 - (4x)^5}{1 - 4x}$$

Ans6:- Write the generating function

$$\begin{aligned} G(x) &= 2 - 2x + 2x^2 - 2x^3 + \dots \\ &= 2(1 - x + x^2 - x^3 + \dots) \end{aligned}$$

this is an infinite generating series

with $a = 1$, $r = -x$

$$1 + r + r^2 + \dots = \frac{1}{1 - r}, \quad |r| < 1$$

$$1 - x + x^2 - \dots = \frac{1}{1 - (-x)} = \frac{1}{1 + x}$$

hence,

$$G(x) = \frac{2}{1 + x}$$

Ans-7:- Given $a_1 + a_2 + a_3 = 17$, $2 \leq a_1 \leq 5$, $3 \leq a_2 \leq 6$,
 $4 \leq a_3 \leq 7$

we have to find the number of integer solutions.

let $a_1 = 2 + x_1$, $a_2 = 3 + x_2$, $a_3 = 4 + x_3$
then

$$(2 + x_1) + (3 + x_2) + (4 + x_3) = 17$$

$$9 + x_1 + x_2 + x_3 = 17$$

$$x_1 + x_2 + x_3 = 8$$

Now from the given bounds:

$$0 \leq x_1 \leq 3, \quad 0 \leq x_2 \leq 3, \quad 0 \leq x_3 \leq 3$$

so we need the number of solutions of

$$x_1 + x_2 + x_3 = 8, \quad 0 \leq x_1, x_2, x_3 \leq 3$$

Since each variable can be at most 3, the only possible way to get sum 8 is

$$3 + 3 + 2$$

Its distinct arrangements are

$$(3, 3, 2) \quad (3, 2, 3) \quad (2, 3, 3)$$

Hence there are 3 solutions.

Now converting back to a_1, a_2, a_3 :

$$(3, 3, 2) \Rightarrow (a_1, a_2, a_3) = (5, 6, 6)$$

$$(3, 2, 3) \Rightarrow (a_1, a_2, a_3) = (5, 5, 7)$$

$$(2, 3, 3) \Rightarrow (a_1, a_2, a_3) = (4, 6, 7)$$

Therefore, the number of solution is 3.

Another Solution:

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So, generating functions for a_1, a_2, a_3 are

$$(x^2 + x^3 + x^4 + x^5), (x^3 + x^4 + x^5 + x^6), (x^4 + x^5 + x^6 + x^7) \text{ respectively}$$

by product rule, total generating function is

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7).$$

Therefore, the number of solutions with the indicated constraints is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7).$$

This follows because we obtain a term equal to x^{17} in the first sum x^{a_1} , a term in the second sum x^{a_2} , and a third sum x^{a_3} , where the exponents a_1, a_2 and a_3 satisfy the equation $a_1 + a_2 + a_3 = 17$ and the given constraints. It can be easily seen that the coefficient of x^{17} in the product is 3. Hence, there are three solutions.